



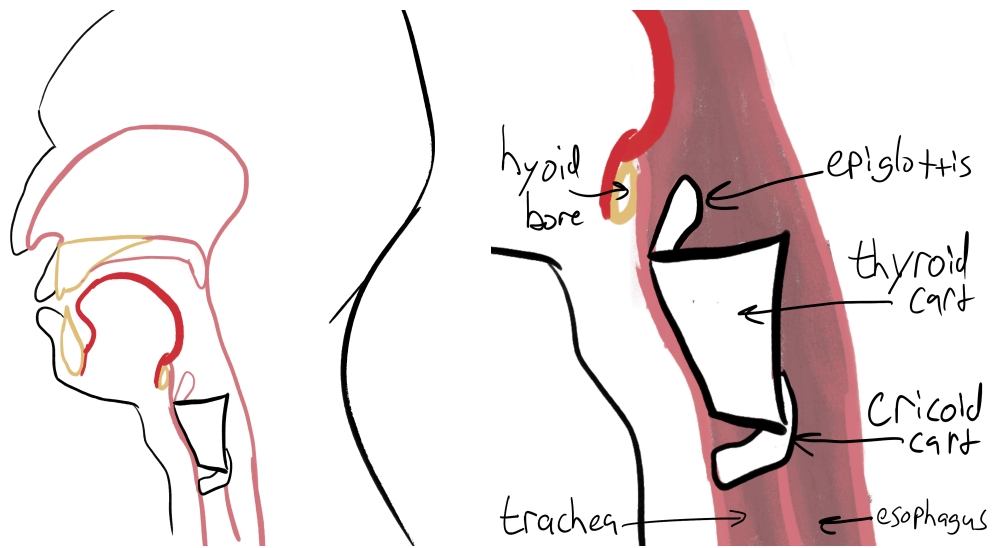
Speech Anatomy 3

The Larynx

Dr. Josef Fruehwald

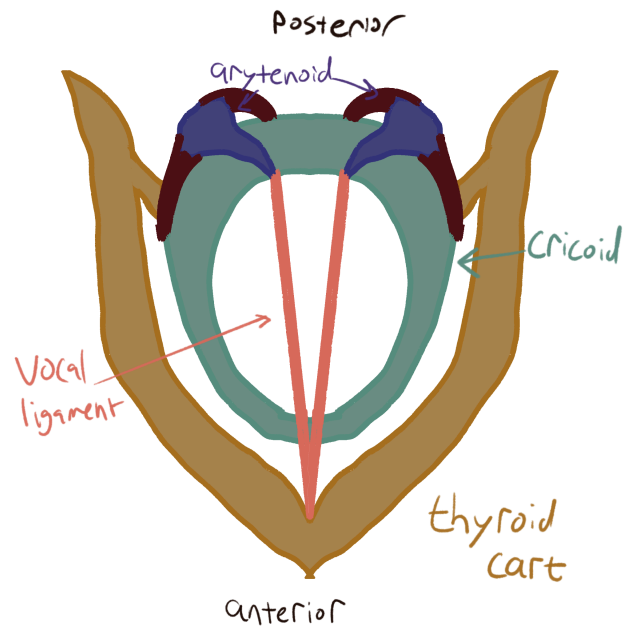
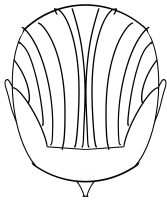
2026-09-01

The Larynx

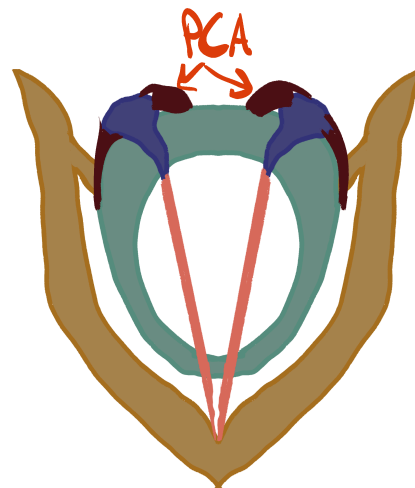
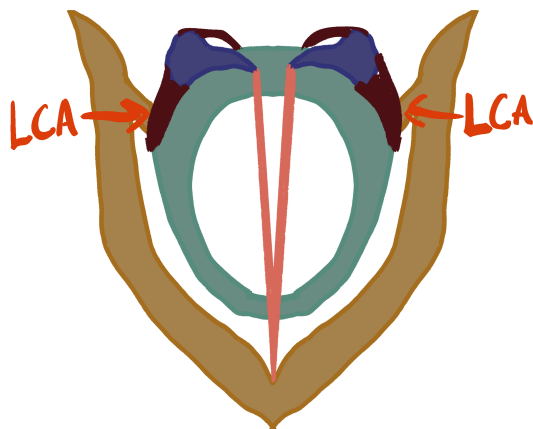




Voicing

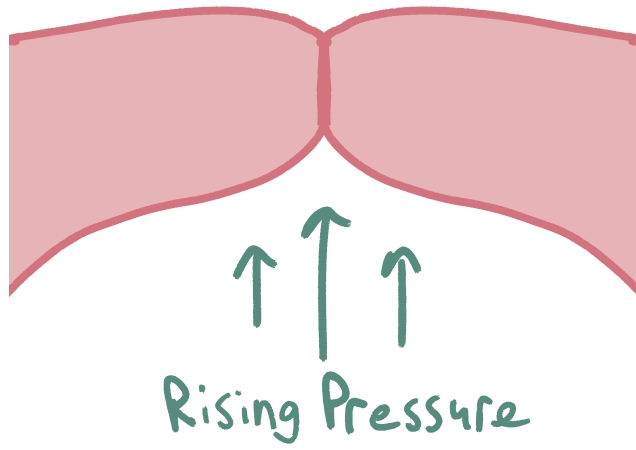
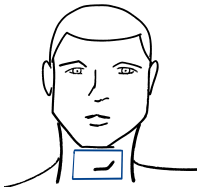


Voicing

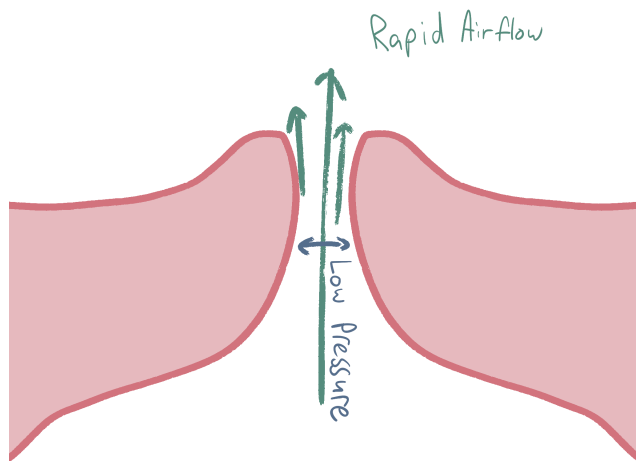
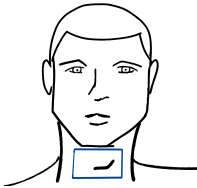




Voicing



Voicing





Voicing

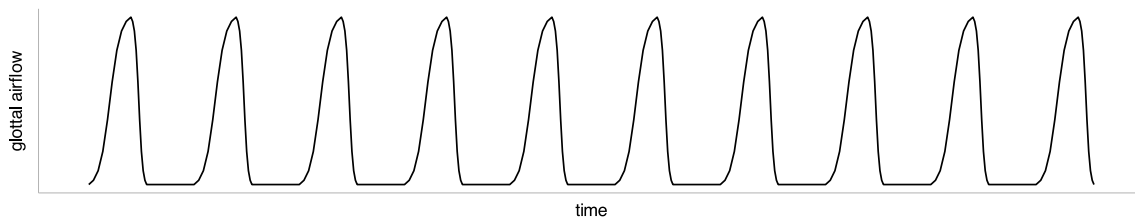
Pitch

Voice Quality

Modal Voice

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
  conflicts to become errors
```



Voice Quality

Modal Voice

- Very “periodic”-glottal pulses at a regular rhythm
- High frequency



Voice Quality

Creaky Voice



Voice Quality

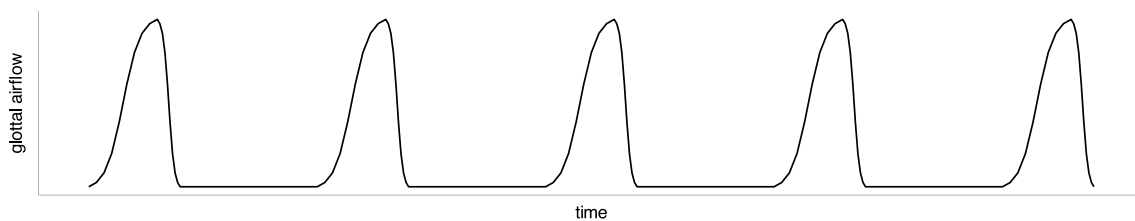
Creaky Voice

- Low frequency
- Some irregular timing between opening phases
- Lower airflow
- Achieved by highly constricted glottis

P. A. Keating, M. Garellek, and J. Kreiman [1]

Voice Quality

Vocal Fry



Voice Quality

Vocal Fry

- Periodic (regular intervals)
- Low frequency
- Achieved by ??

P. A. Keating, M. Garellek, and J. Kreiman [1]



Voice Quality

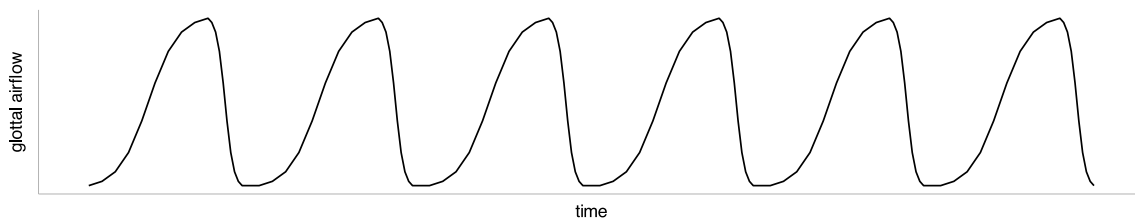
Vocal Fry

Almost everything you've heard about "vocal fry" in popular culture is bunk.

- There's nothing harmful about it.
- People really only complain about women doing it, even though men also do it [2]
- When people complain, they often also don't correctly identify what they're complaining about.

Voice Quality

Breathy voice



Voice Quality

Breathy Voice

- Periodic
- High glottal airflow
- Achieved by laxer vocal fold constriction

References

Bibliography

- [1] P. A. Keating, M. Garellek, and J. Kreiman, "Acoustic properties of different kinds of creaky voice," *Proceedings of the International Congress of Phonetic Science XVIII*, no. 1, 2015, doi: 10.1016/0167-8809(91)90080-H.
- [2] T. Slobe, "This American Creak: Metaphors of Virus, Infection, and Contagion in Girls' Social Networks," in *Paper Presented at AAA 2016*, 2016.